# Four limit cycles in a planar three-reaction mass-action system: a proof of the Boros–Hofbauer conjecture

chemistry-open-problems · 18 October 2026 · verified

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**Abstract.** Boros and Hofbauer conjectured that the planar mass-action system of three reactions $\dot{x} = 1/y - x^a y^b$, $\dot{y} = K(-(d-1) - 1/y + dx^a y^b)$ has, for suitable parameters, an asymptotically stable equilibrium surrounded by four limit cycles, after locating numerically a point where the first three focal values vanish and the fourth is negative; they could not handle $L_2, L_3, L_4$ analytically. We prove the conjecture by two separate computer-assisted arguments. First, we compute the focal values exactly as polynomials over $\mathbb{Q}$ in a square-root-free parametrisation, certify with a Krawczyk operator in ball arithmetic a simple common zero of $L_1, L_2, L_3$ with $L_4 = -2.580126499494\ldots \cdot 10^{-6} < 0$ (rigorous enclosure of radius $10^{-34}$), and conclude by successive perturbations. Second, for explicit decimal parameters, validated Taylor integration of the first-return map brackets four sign changes of the displacement function, giving four nested limit cycles with alternating stability. We also show that this order-four weak focus is the only point of the parameter domain where $L_1 = L_2 = L_3 = 0$.

## 1 The problem

B. Boros, J. Hofbauer, *Oscillations in planar deficiency-one mass-action systems*, J. Dyn. Diff. Equat. 36 (Suppl 1):175–197 (2024), doi:10.1007/s10884-021-10051-z, arXiv:2103.00972 (also PMC10901957), §5.1 ("Four limit cycles"). They consider the three-reaction network (15) with $(a_1, b_1) = (0, 0)$, $(a_2, b_2) = (0, -1)$, $(a_3, b_3) = (a, b)$, $(c_1, d_1) = (0, -1)$, $(c_2, d_2) = (1, -1)$, $(c_3, d_3) = (-1, d)$, $a > 0$, $b > -1$, $d > 0$, $1 + bd > 0$. After scaling the equilibrium to $(1, 1)$ (possible iff $d > 1$) the mass-action ODE (16) becomes

$$\dot{x} = \frac{1}{y} - x^a y^b, \qquad \dot{y} = K\Big( - (d-1) - \frac{1}{y} + d\, x^a y^b \Big), \qquad K > 0. \tag{BH}$$

Verbatim (LaTeX source of arXiv:2103.00972, `sections/5_three_reactions.tex`, §5.1; the same text is in the published version):

> "In this subsection we discuss why we strongly conjecture that there exist parameter values for which the differential equation (16) has at least 4 limit cycles. [...] Set $K = \frac{a}{1+bd}$ to make $\operatorname{tr} J$ equal to zero. The first focal value, $L_1$, is then $L_1 = \frac{\pi a(2a^2 d + a(1+bd) - (1+bd)^2)}{8a\sqrt{(d-1)(1+bd)^3}}$ [...] Thus, there exist $a$, $b$, $d$ such that $L_1 = L_2 = L_3 = 0$ (namely, we numerically find that this happens at $a = \widehat{a} \approx 1.01282$, $b = \widehat{b} \approx 0.65463$, $d = \widehat{d} \approx 3.28862$). Since, again numerically, we

> see that $L_4$ is negative for $\widehat{a}$, $\widehat{b}$, $\widehat{d}$, we conjecture that there exist parameter values for which the unique positive equilibrium of the differential equation (16) is asymptotically stable, and is surrounded by four limit cycles $\Gamma_0$, $\Gamma_1$, $\Gamma_2$, $\Gamma_3$, which are unstable, stable, unstable, stable, respectively. Since the formulas for $L_2$, $L_3$, and $L_4$ get complicated, we cannot handle them analytically. This is why we leave the existence of four limit cycles a conjecture."

## 2 Results

Parameter domain: $\mathcal{P} = \{p = (K, a, b, d) :\ K > 0,\ a > 0,\ b > -1,\ d > 1,\ 1 + bd > 0\}$ (open). For $p \in \mathcal{P}$, $(1, 1)$ is the unique positive equilibrium of (BH), with Jacobian

$$J(p) = \begin{pmatrix} -a & -(1+b) \\ Kda & K(1+bd) \end{pmatrix}, \qquad \det J = Ka(d-1) > 0, \qquad \operatorname{tr} J = -a + K(1+bd).$$

$L_k$ denotes the $k$-th focal value (Lyapunov quantity) at $(1, 1)$ on the hypersurface $\mathcal{T}_0 = \{\operatorname{tr} J = 0\} = \{K = a/(1+bd)\}$, in the normalisation of Boros–Hofbauer (§4.4 below shows that our $L_k$ coincide with theirs on all formulas and values they give).

**Theorem A (four limit cycles).** There is a parameter point $p^* = (K^*, a^*, b^*, d^*) \in \mathcal{T}_0$ with

$$a^* = 1.012821998680314600068208268899 0\ldots, \quad b^* = 0.654625621104238210727259033435 9\ldots,$$
$$d^* = 3.288610439309935787724161356232 6\ldots, \quad K^* = a^*/(1+b^*d^*) = 0.321244360398960809606413563359$$

(truncations of Arb enclosures whose radii are $\leq 2{\cdot}10^{-39}$, see `certificates/certificate_output.json`) such that

1. $L_1(p^*) = L_2(p^*) = L_3(p^*) = 0$;
2. $L_4(p^*) \in [-2.58012649949442448635653476 27 \cdot 10^{-6} \pm 10^{-34}]$, in particular $L_4(p^*) < 0$;
3. the Jacobian matrix $\partial(L_1, L_2, L_3)/\partial(a, b, d)$ at $(a^*, b^*, d^*)$ is nonsingular.

Consequently, every neighbourhood of $p^*$ in $\mathcal{P}$ contains parameters $p$ for which the equilibrium $(1, 1)$ of (BH) is a hyperbolic, asymptotically stable focus surrounded by (at least) four limit cycles $\Gamma_0, \Gamma_1, \Gamma_2, \Gamma_3$ (nested, $\Gamma_0$ innermost), where $\Gamma_0$ and $\Gamma_2$ are repelling (unstable) and $\Gamma_1$, $\Gamma_3$ are attracting (stable). The four cycles can be taken inside any prescribed neighbourhood of $(1, 1)$.

**Corollary (the conjecture of Boros–Hofbauer).** For the network of §5.1 with parameters $(K, a, b, d)$ as produced by Theorem A (or the explicit $p^\sharp$ of Theorem C below) and rate constants $\kappa_1 = d - 1$, $\kappa_2 = 1$, $\kappa_3 = K^{-a}$, the mass-action system (16),

$$\dot{x} = \kappa_2 y^{-1} - \kappa_3 x^a y^b, \qquad \dot{y} = -\kappa_1 - \kappa_2 y^{-1} + d\,\kappa_3 x^a y^b,$$

has the unique positive equilibrium $(K, 1)$, which is asymptotically stable and surrounded by four limit cycles that are unstable, stable, unstable, stable (from the inside out). Indeed, $x = KX$, $y = Y$, $t = K\tau$ transforms it into (BH) (a linear change of coordinates

and a constant positive time rescaling). Hence the conjecture of §5.1 holds, in the precise form stated there.

**Theorem B (uniqueness; bonus).** In the whole domain $\mathcal{P}$, the set $\{\operatorname{tr} J = 0,\ L_1 = L_2 = L_3 = 0\}$ is the single point $p^*$. Consequently, for every $p \in \mathcal{T}_0$ the equilibrium $(1,1)$ is a weak focus of order at most 4 (order exactly 4 only at $p^*$); in particular it is never a centre in this family.

**Theorem C (explicit parameters; second, separate proof).** Let $p^\sharp = (K, a, b, d)$ be the exact decimal fractions

$$\begin{aligned} a &= 1.012822827566885038677856847030028159177026555316278466298 37, \\ b &= 0.654626809470286327709996401918672391083172153706469077547131, \\ d &= 3.288605466176904297183887415659965416086564877981034478548 74, \\ K &= 0.321244556816175423011532938999913745383501015965226302231941 \end{aligned}$$

(`certificates/explicit_params.json`; $\operatorname{tr} J = -1.90 \cdot 10^{-22} < 0$). Then the equilibrium $(1,1)$ of (BH) is a hyperbolic stable focus and the system has (at least) four limit cycles $\Gamma_0, \dots, \Gamma_3$, crossing $\Sigma = \{(1+\rho, 1) : \rho > 0\}$ at $\rho \in (0.002, 0.004)$, $(0.004, 0.006)$, $(0.006, 0.012)$, $(0.012, 0.024)$ respectively, $\Gamma_0, \Gamma_2$ repelling and $\Gamma_1, \Gamma_3$ attracting. The proof (§6.4) uses only validated (interval) Taylor integration of the ODE and planar topology; it does not use the focal values.

*Remark (not used, not re-proved).* With the classical finite-cyclicity theory of a weak focus (Bautin 1952; see e.g. Roussarie 1998, Han–Yu 2012), Theorem B implies that at most four small-amplitude limit cycles can bifurcate from $(1,1)$ in this family, and that for parameters close enough to $p^*$ as in the proof of Theorem A there are exactly four such cycles, all hyperbolic. Our proofs of Theorems A and B do not depend on this remark.

What is proved by what:

| claim | method | status |
|---|---|---|
| exact polynomials $G_k$ $(k \le 4)$ defining $L_k$ | exact rational arithmetic (FLINT) | proved, modulo correctness of a ~250-line program (validated in §4.4, §8; recomputed exactly by the verifier with a different algorithm) |
| $L_1 = L_2 = L_3 = 0$ at a unique point of a box, Jacobian nonsingular | Krawczyk operator in Arb ball arithmetic | proved (computer-assisted) |
| $L_4 < 0$ there | Arb enclosure | proved (computer-assisted) |
| sign lemma, return map, successive perturbations | pen-and-paper (§4.2, §6) | proved |
| Theorem B | exact resultants + Arb root enclosures | proved (computer-assisted) |

| claim | method | status |
|---|---|---|
| Theorem C (explicit rational parameters, four cycles bracketed on $\Sigma$) | validated Taylor integration in Arb + Jordan-curve/Poincaré–Bendixson argument | proved (computer-assisted), independent of the focal-value code |
| location of the cycles to more digits ($\rho \approx 0.004, 0.008, 0.016, 0.032$ on another ray) | 60-digit floating-point Taylor integration | numerical illustration only |

## 3 Set-up and reduction

**3.1 Sqrt-free parametrisation of $\mathcal{T}_0$.** On $\mathcal{T}_0$ the eigenvalues of $J$ are $\pm i\omega$ with $\omega^2 = \det J = a^2(d-1)/(1+bd)$. Put

$$c := 1 + b > 0, \qquad s := \frac{\omega^2}{a^2} = \frac{d-1}{1+bd} > 0, \qquad w := \sqrt{s}, \qquad \rho := 1 - (c-1)s.$$

Then $\rho = (1+b)/(1+bd) > 0$, $d = (1+s)/\rho$, $1 + bd = c/\rho$, $K = a\rho/c$, $K(d-1) = as$, $Kd = a(1+s)/c$. The map $\Psi : (a, c, s) \mapsto (K, a, b, d) = (a\rho/c,\, a,\, c-1,\, (1+s)/\rho)$ is a real-analytic bijection from $\mathcal{D} = \{a > 0,\ c > 0,\ s > 0,\ \rho > 0\}$ onto $\mathcal{T}_0$; the map $(a, b, d) \mapsto (a, c, s)$ has Jacobian determinant $\partial s/\partial d = (1+b)/(1+bd)^2 > 0$.

**3.2 Normal coordinates.** Let $u = x - 1$, $v = y - 1$, $I(v) = (1+v)^{-1}$, $E(u, v) = (1+u)^a(1+v)^b$, $f = I - E$. Using $\dot{y} = K[(d-1)(I-1) - df]$, the linear change and time rescaling

$$u = cX, \qquad v = a(wY - X), \qquad \tau = \omega t = aw\, t$$

transform (BH) (for $p = \Psi(a, c, s)$) into

$$X' = -Y + P(X, Y), \qquad Y' = X + Q(X, Y), \qquad P = \frac{N_f}{acw}, \quad Q = \frac{cN_I - N_f}{ac}, \tag{N}$$

where $N_I, N_f$ are the parts of degree $\geq 2$ in $(X, Y)$ of $I$ and $f$ (direct check: the linear parts are $-Y$ and $X$). Every Taylor coefficient of $N_I, N_f$ is a polynomial in $(a, c, w)$ because $(1+u)^a(1+v)^{c-1} = \sum_{i,j} \binom{a}{i}\binom{c-1}{j} u^i v^j$. The series converge for $|u|, |v| < 1$, so (N) is real-analytic near the origin, jointly in the parameters.

## 4 Focal values

**4.1 Algorithm (complex Lyapunov-function method).** Write $z = X + iY$; (N) reads $z' = iz + R(z, \bar{z})$, $R = P + iQ = \sum_{p+q\geq 2} R_{pq} z^p \bar{z}^q$. We look for a real polynomial $F = z\bar{z} + \sum_{3\leq p+q\leq N} f_{pq} z^p \bar{z}^q$ ($f_{qp} = \overline{f_{pq}}$). Along (N),

$$\dot{F} = \sum i(p-q) f_{pq} z^p \bar{z}^q + \sum f_{rs}(r z^{r-1} \bar{z}^s R + s z^r \bar{z}^{s-1} \bar{R}).$$

Let $S_{pq}$ be the coefficient of $z^p\bar z^q$ in the second sum computed with the $f$ of degree $< p+q$ (only those contribute, because $R = O(|z|^2)$). Degree by degree, $n = 3, 4, \ldots$:

$$f_{pq} = \frac{i\,S_{pq}}{p-q}\ (p \neq q), \qquad V_k := S_{k+1,k+1} \in \mathbb{R}\ (n = 2k+2), \qquad f_{k+1,k+1} := 0.$$

Then, for every $j \geq 1$, the polynomial $F_j$ built up to degree $2j+2$ satisfies

$$\dot F_j = \sum_{k=1}^{j} V_k\,|z|^{2k+2} + O(|z|^{2j+3}). \tag{4.1}$$

(The terms of degree $\leq 2j+2$ of $\dot F_j$ are exactly the ones cancelled or recorded by the recursion; all remaining terms have degree $\geq 2j+3$ because $F_j$ is a polynomial and $R = O(|z|^2)$ is analytic.) The choice $f_{k+1,k+1} = 0$ is a convention; Lemma 4.2 holds for any choice.

**4.2 Sign lemma.** *Let $p \in \mathcal{T}_0$ and suppose $V_1(p) = \cdots = V_{j-1}(p) = 0 \neq V_j(p)$. Let $\Sigma$ be any ray issuing from $(1,1)$ and transversal to the flow near $(1,1)$ (e.g. $\{(1+\rho, 1) : \rho > 0\}$), and $\delta(\rho) = \mathcal{P}(\rho) - \rho$ the displacement function of the first-return map on $\Sigma$ (§6.1). Then $\operatorname{sign}\delta(\rho) = \operatorname{sign} V_j(p)$ for all sufficiently small $\rho > 0$.*

*Proof.* By (4.1), $\dot F_j = V_j|z|^{2j+2}(1 + O(|z|))$, so $\dot F_j$ has the sign of $V_j$ on a punctured disc $0 < |z| < r_0$. Since $F_j = |z|^2 + O(|z|^3)$, $F_j$ is positive definite and its restriction to the ray $\Sigma$ is strictly increasing in $\rho$ for small $\rho$ ($F_j = q\rho^2 + O(\rho^3)$, $q > 0$, in any linear coordinates). For $\rho$ small, the orbit from $\rho$ stays in the punctured disc during one turn (§6.1), so $F_j(\mathcal{P}(\rho)) - F_j(\rho) = \int \dot F_j\,dt$ has the sign of $V_j$, hence so does $\mathcal{P}(\rho) - \rho$. □

(When $V_1 = \cdots = V_{j-1} = 0$, (4.1) gives $d(|z|^2)/d\tau \approx V_j|z|^{2j+2}$, i.e. $\delta(r) = \pi V_j r^{2j+1} + O(r^{2j+2})$ in the coordinate $r = |z|$ of (N). Using the unscaled coordinate $X = u$ instead of $u = cX$ gives the normalisation of the paper, $L_j = \pi V_j^{(u)}$; see §4.3.)

**4.3 Exact polynomials.** Put $D = acw$ and $\tilde R := D\,R = N_f + iw(cN_I - N_f)$ (polynomial coefficients). Running the recursion on $\tilde f_{pq} := D^{\,p+q-2} f_{pq}$ gives the polynomial recursion

$$\tilde S_n = \sum_{m=2}^{n-1} D^{\,n-1-m}\big[\tilde F_m \otimes \tilde R_{n-m+1}\big], \qquad \tilde f_{pq} = \frac{i\tilde S_{pq}}{p-q}\,, \qquad \tilde V_k = \tilde S_{k+1,k+1} = D^{2k}V_k,$$

so that $\tilde V_k \in \mathbb{Q}[a, c, w]$. The program `scripts/focal_exact.py` (python-flint `fmpq_mpoly`, complex numbers as pairs of polynomials) computes $\tilde V_1, \ldots, \tilde V_4$ in 0.1 s and checks along the way that every $\tilde S_{k+1,k+1}$ is real and that $\tilde S_{qp} = \overline{\tilde S_{pq}}$. Every $\tilde V_k$ is $w$ times a polynomial in $(a, c, w^2)$; we set

$$G_k(a, c, s) := \tilde V_k(a, c, \sqrt{s})/\sqrt{s} \in \mathbb{Q}[a, c, s], \qquad L_k = \pi\,V_k^{(u)} = \frac{\pi\,G_k(a, c, s)}{a^{2k}c^{4k}s^{\,k-1/2}}\,,$$

where $V_k^{(u)}$ is the Lyapunov quantity in the coordinates $X = u$ (unscaled), i.e. $V_k^{(u)} = c^{-2k}V_k$. On $\mathcal{D}$, $\operatorname{sign} L_k = \operatorname{sign} V_k = \operatorname{sign} G_k$, so Lemma 4.2 holds with $G_k$ in place of $V_k$. Factorisations (exact):

$$G_1 = -\tfrac{1}{8}\,a^2c^2\,g_1, \quad G_2 = \tfrac{1}{1152}\,a^4c^4\,g_2, \quad G_3 = -\tfrac{1}{442368}\,a^6c^6\,g_3, \quad G_4 = \tfrac{1}{530841600}\,a^8c^8\,g_4,$$

with $g_k \in \mathbb{Z}[a,c,s]$ irreducible, of degrees $(2k, 2k, 3k-1)$ in $(a,c,s)$ and 9, 48, 136, 296 terms. Explicitly

$$-g_1 = 2a^2(1+s)\rho + ac\rho - c^2, \qquad \rho = 1 + s - cs,$$

and $g_2, g_3$ are listed in Appendix A; all four are stored in `certificates/G_k.json`.

**4.4 Validation against the paper.** (i) Exact identity (`crosscheck.py`): $L_1 = \pi G_1/(a^2c^4\sqrt{s}) = \pi[2a^2(1+s)\rho + ac\rho - c^2]/(8c^2\sqrt{s})$ equals the paper's $\pi a(2a^2d + a(1+bd) - (1+bd)^2)/(8a\sqrt{(d-1)(1+bd)^3})$ after $d = (1+s)/\rho$, $1+bd = c/\rho$. (ii) The generic floating-point implementation `focal_mp.py` (separate code path: numerical Taylor expansion of arbitrary generalized-monomial systems, numerical normalisation with $\omega = \sqrt{\det J}$, no reparametrisation) reproduces **exactly** (ratio 1.000… to 15+ digits) every focal value printed in the paper: $L_1$ of §3.2, §4.1 and §5.1 (two generic points each), $L_2$ of §4.1 along $\{L_1 = 0\}$ (two points), $L_2 \approx 0.01293$ of §3.2, and $L_3 = -\frac{625\pi}{110592}\sqrt{7/2}$ of §4.1. (iii) The exact $L_k$ agree with `focal_mp.py` at four parameter points to relative precision $10^{-55}$ for $k = 1..4$. (iv) The paper's checkpoints: at $(a,d) = (1, 165/49)$ (then $b = 7/11$, $s = 58/77$) we get exactly $L_1 = L_2 = 0$ and $L_3 < 0$ (exact rational sign); at $(a,d) = ((1+\sqrt{3961})/60, 3)$, $L_1 = L_2 = 0$ (to 50 digits) and $L_3 > 0$ — as stated in the paper.

## 5 The interval certificate (`scripts/certify.py`)

**Krawczyk's theorem** (Krawczyk 1969; Moore 1977; Neumaier 1990, §5.1). Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be $C^1$ on a box $X$, $m \in X$, $C$ a real $n \times n$ matrix and $\mathbf{J}$ an interval matrix containing $G'(x)$ for all $x \in X$. If $\mathcal{K}(X) = m - CG(m) + (I - C\mathbf{J})(X - m) \subset \operatorname{int} X$, then $G$ has exactly one zero in $X$. Moreover, if $\|I - C\mathbf{J}\|_\infty < 1$, every matrix in $\mathbf{J}$ is nonsingular (Neumann series for $CA$).

Data (`certificates/box.json`): variables $(a,c,s)$, centre $m$ = three rationals with denominator $\le 10^{60}$ (Newton iterate at 150 digits), radius $r = 10^{-40}$, preconditioner $C$ = rational approximation of $G'(m)^{-1}$. Computations in Arb (python-flint 0.9.0, 512-bit midpoint–radius balls with outward rounding); $G = (G_1, G_2, G_3)$, $\mathbf{J}$ = exact derivative polynomials evaluated on $X$.

| check | result |
|---|---|
| C0 recomputed $G_k$ = stored $G_k$; C0′ Arb balls contain the exact box | PASS |
| C1 on $X$: $a > 0$, $c > 0$, $s > 0$, $\rho > 0$, $d > 1$, $1 + bd > 0$, $K > 0$ | PASS |
| C2 $\mathcal{K}(X) \subset \operatorname{int} X$: $\max_i \lvert\mathcal{K}_i - m_i\rvert \le 3.62 \cdot 10^{-61} \ll r = 10^{-40}$ | PASS |
| C3 $\|I - C\mathbf{J}\|_\infty \le 1.37 \cdot 10^{-32} < 1$; $\det \mathbf{J} \in [-0.208483123515217 \pm 4 \cdot 10^{-16}]$ | PASS |

| check | result |
|---|---|
| C4<br>$G_4(X) \subset [-9.3543144375364141858217679 70 \cdot 10^{-4} \pm 2 \cdot 10^{-32}] < 0$;<br>$L_4(X) \subset [-2.5801264994944244863565347627 \cdot 10^{-6} \pm 10^{-34}]$ | PASS |
| C5 $\det \partial(G_1, G_2, G_3)/\partial(a, b, d) \in [-0.03470367143 \pm 4 \cdot 10^{-12}]$ | PASS |

Let $\mu^* \in X$ be the unique zero and $p^* = \Psi(\mu^*)$. At $\mu^*$, $G_1 = G_2 = G_3 = 0$, so $L_1 = L_2 = L_3 = 0$; $L_4(p^*) < 0$ by C4; and since $L_k = \pi G_k/(\text{positive})$, at a common zero $\partial L_k = \pi\, \partial G_k/(\text{positive})$, so C5 gives item 3 of Theorem A. The enclosures of $(a^*, b^*, d^*, K^*)$ quoted in §2 are the Arb images of $X$ (`certificates/certificate_output.json`). The paper's values $a \approx 1.01282$, $b \approx 0.65463$, $d \approx 3.28862$ agree with ours except in the last digit of $d$ ($d^* = 3.288610\ldots$), which reflects their numerical accuracy.

## 6 The bifurcation argument (proof of Theorem A) and validated integration (proof of Theorem C)

Let $\Sigma = \{(1+\rho, 1) : \rho > 0\}$. On $\Sigma$, $\dot{y} = Kd((1+\rho)^a - 1) > 0$: $\Sigma$ is transversal.

**6.1 Lemma (return map).** Let $p_0 \in \mathcal{T}_0$. There are a compact neighbourhood $U \subset \mathcal{P}$ of $p_0$ and $\bar{\rho} > 0$ such that for all $p \in U$ and $\rho \in (0, \bar{\rho}]$ the forward orbit of $(1+\rho, 1)$ returns to $\Sigma$ after one turn around $(1,1)$, at $(1 + \mathcal{P}(\rho, p), 1)$; the map $\mathcal{P}$ is continuous on $(0, \bar{\rho}] \times U$, real-analytic in $\rho$, and strictly increasing in $\rho$. Each zero $\rho_0 \in (0, \bar{\rho}]$ of $\delta(\cdot, p) = \mathcal{P}(\cdot, p) - \mathrm{id}$ gives a periodic orbit through $(1+\rho_0, 1)$ surrounding $(1,1)$; distinct zeros give distinct periodic orbits, and an isolated zero gives an isolated periodic orbit, i.e. a limit cycle.

*Proof (standard).* Let $T$ bring $J(p_0)$ to the rotation form and use polar coordinates $(r, \theta)$ for $\xi = T(u, v)$. Since $J(p) \to J(p_0)$ as $p \to p_0$ and the nonlinear terms are $O(r^2)$ uniformly on a compact parameter set, there are $U$, $r_0 > 0$ with $\dot{\theta} \geq \omega_0/2$ and $|\dot{r}| \leq Lr$ for $r \leq r_0$, $p \in U$. For $r(0) \leq \bar{r} := r_0 e^{-4\pi L/\omega_0}$ the orbit stays in $r \leq r_0$ up to time $4\pi/\omega_0$, during which $\theta$ increases by at least $2\pi$; hence it meets the ray $\Sigma$ (a ray in the $\xi$-plane too) again. The crossings are transversal, so the implicit function theorem gives continuity in $(\rho, p)$ and analyticity in $\rho$. Monotonicity follows from uniqueness of solutions (orbits do not cross). A periodic orbit in $\{r \leq r_0\}$ has winding number $\pm 1$ around $(1,1)$ and $\theta$ is strictly increasing along it, so it meets $\Sigma$ exactly once per period; a periodic orbit near one through $\rho_0$ meets $\Sigma$ near $\rho_0$, hence is the same orbit if $\rho_0$ is an isolated zero. □

**6.2 Lemma (trace).** If $p \in U$ and $\operatorname{tr} J(p) < 0$, then $\delta(\rho, p) < 0$ for all small $\rho > 0$.
*Proof.* $(1,1)$ is a hyperbolic sink ($\det J > 0$), attracting a neighbourhood. If $\delta(\rho, p) = 0$ there would be a periodic orbit in that neighbourhood; if $\delta(\rho, p) > 0$, the iterates $\mathcal{P}^n(\rho)$ would increase (monotonicity) and not tend to 0. Both contradict attraction. □

**6.3 Proof of Theorem A.** Let $G = (G_1, G_2, G_3)$. By C2–C3, $G(\mu^*) = 0$ and $G'(\mu^*)$ is invertible, so (inverse function theorem) $G$ maps a neighbourhood $N$ of $\mu^*$ diffeomorphically onto a neighbourhood $W$ of 0; write $\Phi = \Psi \circ (G|_N)^{-1} : W \to \mathcal{T}_0$,

continuous with $\Phi(0) = p^*$. Fix $U, \bar\rho$ from Lemma 6.1 for $p_0 = p^*$ and shrink $W$ so that $\Phi(W) \subset U$. Given any neighbourhood $\mathcal{V}$ of $p^*$ and $\varepsilon > 0$, we construct $p_4 \in \mathcal{V}$ and $0 < \rho_5 < \rho_4 < \rho_3 < \rho_2 < \rho_1 < \varepsilon$ with

$$\delta(\rho_5, p_4) < 0 < \delta(\rho_4, p_4), \quad \delta(\rho_3, p_4) < 0 < \delta(\rho_2, p_4), \quad \delta(\rho_1, p_4) < 0.$$

- Step 0. At $p^*$: $G_1 = G_2 = G_3 = 0$, $G_4 < 0$. Lemma 4.2 ($j = 4$) gives $\rho_1 \in (0, \min(\bar\rho, \varepsilon))$ with $\delta(\rho_1, p^*) < 0$.
- Step 1. By continuity of $\delta(\rho_1, \cdot)$ choose $t_3 > 0$ so small that $p_1 := \Phi(0, 0, t_3) \in \mathcal{V}$ still has $\delta(\rho_1, p_1) < 0$. At $p_1$: $G_1 = G_2 = 0 < G_3$, so by Lemma 4.2 there is $\rho_2 < \rho_1$ with $\delta(\rho_2, p_1) > 0$.
- Step 2. Choose $t_2 > 0$ so small that $p_2 := \Phi(0, -t_2, t_3) \in \mathcal{V}$ keeps the signs at $\rho_1, \rho_2$. At $p_2$: $G_1 = 0 > G_2$: choose $\rho_3 < \rho_2$ with $\delta(\rho_3, p_2) < 0$.
- Step 3. Choose $t_1 > 0$ so small that $p_3 := \Phi(t_1, -t_2, t_3) \in \mathcal{V}$ keeps the signs at $\rho_1, \rho_2, \rho_3$. At $p_3$: $G_1 > 0$: choose $\rho_4 < \rho_3$ with $\delta(\rho_4, p_3) > 0$.
- Step 4. Let $p_4 = p_3 - (\eta, 0, 0, 0)$ (decrease $K$), $\eta > 0$ so small that $p_4 \in \mathcal{V} \cap U$ keeps the signs at $\rho_1, \dots, \rho_4$. Now $\operatorname{tr} J(p_4) = -\eta(1 + bd) < 0$; Lemma 6.2 gives $\rho_5 < \rho_4$ with $\delta(\rho_5, p_4) < 0$.

(All the $\Phi(\cdot)$ points lie on $\mathcal{T}_0$, where Lemma 4.2 applies; continuity of $\Phi$ makes each new point as close to the previous one as needed.) Since $\delta(\cdot, p_4)$ is analytic on $(0, \bar\rho]$ and not identically zero, it has finitely many zeros in $[\rho_5, \rho_1]$. On each of the four intervals $(\rho_5, \rho_4), (\rho_4, \rho_3), (\rho_3, \rho_2), (\rho_2, \rho_1)$ it changes sign; take the zeros $\zeta_0 < \zeta_1 < \zeta_2 < \zeta_3$, one in each interval, at which the sign changes from $-$ to $+$ ($\zeta_0, \zeta_2$) resp. from $+$ to $-$ ($\zeta_1, \zeta_3$) when $\rho$ increases (they exist because the finitely many gaps between consecutive zeros carry constant signs that start with $-$, resp. $+$, and end with the opposite sign). Let $\Gamma_i$ be the periodic orbit through $(1 + \zeta_i, 1)$ — a limit cycle by Lemma 6.1. If $\delta < 0$ just inside and $\delta > 0$ just outside $\zeta_0$, orbits near $\Gamma_0$ move away from it on both sides ($\mathcal{P}$ is increasing and fixes $\zeta_0$): $\Gamma_0$ is repelling; likewise $\Gamma_2$. At $\zeta_1, \zeta_3$ the signs are reversed and the iterates converge monotonically to $\zeta_i$ from both sides: $\Gamma_1, \Gamma_3$ are attracting. The equilibrium is a hyperbolic sink ($\operatorname{tr} J < 0 < \det J$). The $\Gamma_i$ are nested Jordan curves around $(1, 1)$ (each meets $\Sigma$ once, at increasing $\zeta_i$) and lie in any prescribed neighbourhood of $(1, 1)$ if $\varepsilon$ is small. □

### 6.4 Proof of Theorem C (`scripts/validated_cycles.py`).

*Validated integrator.* With $E = x^a y^b$, $I = 1/y$, $J = 1/x$, (BH) lifts to the polynomial system $x' = I - E$, $y' = K(-(d-1) - I + dE)$, $E' = E(aJx' + bIy')$, $I' = -I^2 y'$, $J' = -J^2 x'$; from consistent initial data its solution projects onto that of (BH) (indeed $q = Iy$ and $p = Jx$ satisfy $q' = Iy'(1 - q)$, $p' = Jx'(1 - p)$, so $q \equiv p \equiv 1$ and $y, x$ stay positive; then $E' = E(ax'/x + by'/y)$ gives $E \equiv x^a y^b$). One step of length $h$ from a ball enclosure $Z_k$ of the state: (i) find a box $B$ with $Z_k + [0, h]\, F(B) \subset \operatorname{int} B$ (then every solution starting in $Z_k$ exists on $[0, h]$ and stays in $B$: if it left $B$ at a first time $t_1 \le h$, then $z(t_1) \in z_0 + t_1 \operatorname{conv} F(B) \subset \operatorname{int} B$, a contradiction); (ii) Taylor coefficients $z_j(Z_k)$, $j < N$, by the standard recurrences in Arb; (iii) Lagrange remainder $z_N(B)\tau^N$ (the $N$-th Taylor coefficient is a polynomial function of the state, and the intermediate state lies in $B$). Hence $Z(t_k + \tau) \in \sum_{j<N} z_j(Z_k)\tau^j + z_N(B)\tau^N$ for $\tau \in [0, h]$. Parameters: $N = 36$, $h = 0.05$ (146 steps per turn), 320-bit balls. The enclosures of $\delta$ do not change (15 digits) with $(N, h) = (28, 0.03)$ or $(44, 0.08)$.

*Crossing bookkeeping.* $\Sigma$ is transversal ($\dot{y} = Kd((1+\rho)^a - 1) > 0$). In every step whose box $B$ meets the line $y = 1$, the script requires either ($x < 1$ and $\dot{y} < 0$ on $B$) or ($x > 1$ and $\dot{y} > 0$ on $B$), so $y$ is strictly monotone there and crossings are counted exactly: for every test point the orbit leaves $\Sigma$ upwards, crosses the left half-line $\{x < 1, y = 1\}$ exactly once (downwards), and the first return to $\Sigma$ is enclosed by bisection of the crossing time on the Taylor model ($2^{-140}$ relative width), giving $\mathcal{P}(\rho)$ to about $10^{-38}$.

*Certified values* (`certificates/validated_cycles_output.json`):

| $\rho$ | $\delta(\rho) = \mathcal{P}(\rho) - \rho$ |
|---|---|
| 0.002 | $[-4.60438730999 \cdot 10^{-25} \pm 6.2 \cdot 10^{-38}]$ |
| 0.004 | $[+1.27356613600 \cdot 10^{-24} \pm 1.9 \cdot 10^{-36}]$ |
| 0.006 | $[-3.54914581811 \cdot 10^{-24} \pm 4.5 \cdot 10^{-36}]$ |
| 0.012 | $[+1.41126309912 \cdot 10^{-22} \pm 4.5 \cdot 10^{-34}]$ |
| 0.024 | $[-3.02911287653 \cdot 10^{-20} \pm 4.8 \cdot 10^{-32}]$ |

(radii as printed by Arb with 12 significant digits, i.e. including the display rounding).

(also $\operatorname{tr} J < 0 < \det J$, $(\operatorname{tr} J)^2 < 4 \det J$ in exact rational arithmetic).

*Topology.* Let $\rho_1 = 0.024$, $q_1 = (1 + \rho_1, 1)$, $\gamma$ the orbit arc from $q_1$ to its first return $q_1' = (1 + \mathcal{P}(\rho_1), 1)$, $\mathcal{P}(\rho_1) < \rho_1$. Use polar coordinates $(r, \theta)$ around $(1, 1)$ in the plane $(u, v) = (x - 1, y - 1)$, so that $\Sigma = \{\theta = 0\}$. The script certifies $u\dot{v} - v\dot{u} > 0$ on every enclosure box along $\gamma$, i.e. $\theta$ is strictly increasing along $\gamma$; with the crossing bookkeeping ($\gamma$ meets $\{\theta = \pi\}$ once and $\{\theta = 0\}$ only at its end points), $\theta$ increases by exactly $2\pi$, so $\gamma$ is a polar graph $r = R(\theta)$, $0 \le \theta \le 2\pi$, with $R(0) = \rho_1 > R(2\pi) = \mathcal{P}(\rho_1)$. Hence $C = \gamma \cup [q_1', q_1]$ is a Jordan curve bounding the star-shaped region $D = \{r \le R(\theta),\ 0 < \theta < 2\pi\} \cup \{\theta = 0,\ r \le \rho_1\}$, which contains $(1, 1)$ and the whole segment $\{(1 + \rho, 1) : 0 < \rho \le \rho_1\}$. A point of the open segment $(q_1', q_1)$ has $R(2\pi^-) = \mathcal{P}(\rho_1) < r < \rho_1 = R(0^+)$: the flow, which crosses $\Sigma$ with increasing $\theta$, enters $D$ there; orbits cannot cross $\gamma$. So $D$ is positively invariant. For every $\rho \in (0, \rho_1]$ the orbit of $(1+\rho, 1)$ stays in $D$, where $(1, 1)$ is the only equilibrium (it is the unique positive equilibrium); by Poincaré–Bendixson its $\omega$-limit set is $(1, 1)$ (a focus: the orbit spirals and crosses $\Sigma$) or a periodic orbit surrounding $(1, 1)$ (which meets $\Sigma$ transversally, so the orbit crosses $\Sigma$ near it); a homoclinic loop is impossible since $(1, 1)$ is a sink. Thus the first return map $\mathcal{P}$ is defined on $(0, \rho_1]$, and by transversality it is continuous and real-analytic there. By the table, $\delta$ changes sign on each of the four intervals; as in §6.3 (finitely many zeros, choice of sign-changing zeros, monotonicity of $\mathcal{P}$) we get four limit cycles with the stated stabilities. □

The parameters $p^\sharp$ were found with the focal values (targets $(G_1, G_2, G_3)$ prescribed so that the truncated displacement has roots at $0.004, 0.008, 0.016, 0.032$ on the ray $Y = 0$, see §8.2), but the proof of Theorem C does not depend on how they were found.

**Remarks.** (a) Only the network parameters $(a, b, d)$ and $K = \bar{x}/\bar{y}$ are varied, exactly as in the paper's §§4.1 and 5.1; the conjecture is about "parameter values" of (16), including the exponents. (For a fixed network the rate constants enter the scaled system (BH) only through $K$ — the scaled constants $\overline{\kappa}_i$ are fixed up to the factor $\lambda$, which is absorbed by a time rescaling — so a fixed network only gives a one-parameter family and

the network parameters must be varied, as the paper does; compare arXiv:2406.13451 for a general statement in the quadratic setting.) (b) The perturbation directions are the paper's: first $L_3$, then $L_2$, $L_1$, finally the trace via $K$.

## 7 Proof of Theorem B (`scripts/uniqueness.py`)

On $\mathcal{D}$, $G_1 = 0 \iff 2a^2(1+s)\rho + ac\rho - c^2 = 0$. With $t := a/c > 0$ this reads $\rho\, t(2t(1+s)+1) = 1$, i.e. $\rho = 1/(tQ)$, $Q = 2t(1+s)+1$, and $c = (1+s-\rho)/s$. Hence $(t,s) \mapsto (a,c,s) = (tc,c,s)$ is a bijection from $\{t>0,\ s>0,\ 2t(1+s)>1\}$ (the last condition is $c>0$) onto $\{G_1 = 0\} \cap \mathcal{D}$. Substituting and clearing the denominator $(stQ)^m$:

- $G_1$ becomes identically 0 (check);
- $H_2 = \text{const} \cdot (\text{powers of } s,\ s+1,\ t,\ ts+t+1,\ 2ts+2t-1) \cdot h_2(t,s)$, and the same for $H_3$ with $h_3$; all the listed small factors are nonzero on the $(t,s)$-domain ($2t(1+s)-1>0 \iff c>0$); $h_2$, $h_3$ have degrees $(6,4)$, $(10,8)$ and 24, 68 terms.
- $R_t = \mathrm{Res}_s(h_2,h_3)$ (degree 58) factors as $t^{32}(t+1)^9(2t-1)^4(2t+3)(3t-1)\,p_{11}(t)$ with $p_{11} = 628224t^{11} + 1764800t^{10} - 1694592t^9 - 15308848t^8 - 25081168t^7 + 10994160t^6 + 36659948t^5 - 2189226t^4 - 16683601t^3 + 3894133t^2 - 2248520t + 1436225$; $R_s = \mathrm{Res}_t(h_2,h_3)$ has degree 56.
- Any common zero $(t_0,s_0)$ satisfies $R_t(t_0) = R_s(s_0) = 0$. All complex roots of both resultants are enclosed in Arb balls; the candidates with possibly positive real part and possibly zero imaginary part are $t \in \{1/3,\ 0.61212,\ 0.61956,\ 0.74045,\ 2.73230,\ 1/2\}$ and $s \in \{0.11924,\ 0.72590\}$. For all pairs but one, the Arb enclosure of $h_2$ or $h_3$ excludes 0. The surviving pair, mapped back to $(a,c,s)$, lies inside the Krawczyk box $X$ of §5, where the zero is unique.

Hence $\{G_1 = G_2 = G_3 = 0\} \cap \mathcal{D} = \{\mu^*\}$, which is Theorem B (the order statement follows from $L_4(p^*) \neq 0$). (PARI/GP gives the same picture: over $\mathbb{Q}(\theta)$, $p_{11}(\theta) = 0$, $\gcd(h_2,h_3)$ is linear in $s$; the real roots $\theta \approx 0.6196, 0.7405, 2.7323$ give $s<0$.)

## 8 Non-rigorous cross-checks (not part of the proof)

1. **Separate method** (`polar_check.py`): the coefficients $u_k(2\pi)$ of the displacement map, obtained by integrating the triangular ODE system of the polar expansion $dr/d\theta = \sum C_k(\theta) r^k$ with DOP853 (tolerance $10^{-13}$), agree with $\pi V_k$: $u_3 = \pi V_1$ at a generic point (rel. error $6 \cdot 10^{-15}$); on $\{L_1 = 0\}$: $u_3 \approx 0$, $u_5 = \pi V_2$ ($10^{-11}$); on $\{L_1 = L_2 = 0\}$: $u_5 \approx 0$, $u_7 = \pi V_3$ ($8 \cdot 10^{-11}$); at $p^*$: $u_2, \ldots, u_8 = O(10^{-14})$ and $u_9(2\pi) = -2.5801263 \cdot 10^{-6} = \pi V_4$ ($5 \cdot 10^{-8}$).
2. **Explicit four cycles** (`illustrate_cycles.py`, 60-digit arithmetic, Taylor integrator of order 40 on the polynomial lift $(x, y, x^a y^b, 1/y, 1/x)$): targets chosen so that the truncated displacement has roots $0.004, 0.008, 0.016, 0.032$; parameters $a = 1.01282282756688503867785684703002 8\ldots$, $b = 0.65462680947028632770999640191867 2\ldots$, $d = 3.28860546617690429718388741565996 5\ldots$, $K = a/(1+bd) - 6.03 \cdot 10^{-23}$ (full 60 digits printed by the script). Displacements on the ray $\{x = 1+\rho,\ y = 1-\rho a/(1+b)\}$: $\delta(0.002) = -9.56 \cdot 10^{-25}$, $\delta(0.006) = +1.89 \cdot 10^{-24}$, $\delta(0.012) = -3.12 \cdot 10^{-23}$, $\delta(0.024) = +2.41 \cdot 10^{-21}$, $\delta(0.045) = -7.01 \cdot 10^{-19}$ (truncated predictions within 0.1–15%); bisection locates the four cycles at $\rho \approx 0.004006,\ 0.008019,\ 0.01614,\ 0.03230$. This is floating-point evidence only (no validated integration).
3. The paper's checkpoints and formulas (§4.4).

## 9 Trust base, limitations, what a verifier should check first

- **Two separate proofs.** Theorem A rests on the focal-value code + Krawczyk + the bifurcation argument; Theorem C rests only on the validated integrator (~150 lines, `validated_cycles.py`) + Arb + planar topology. Either one alone settles the conjecture. A verifier should check in Theorem C: the a priori enclosure test and the remainder term (§6.4), the crossing bookkeeping (no missed crossing of $\Sigma$), and the Jordan-curve/positive-invariance argument.
- **Correctness of the focal-value code.** The certificate checks the polynomials $G_k$ that `focal_exact.py` produces; it does not prove that these are the Lyapunov quantities. Evidence: the derivation in §3–4 (short, checkable by hand), the exact match with the paper's closed-form $L_1$, the match of a separate generic implementation with every focal value printed in the paper ($L_1$, $L_2$, $L_3$ of three different systems), and a separate method (polar expansion) at the degenerate points, including $u_9(2\pi) = \pi V_4$ at $p^*$. A verifier should re-derive $\tilde{R} = N_f + iw(cN_I - N_f)$ and, ideally, recompute $L_1, \dots, L_4$ at $p^*$ with a separate CAS/algorithm (e.g. Singular/Maple normal form). *Done in the verification* (`verification.md`, scripts in `verification/`): the focal values were recomputed exactly with a different algorithm (real-coordinate Lyapunov function, homogeneous linear algebra with the circle-average normalisation, separately derived vector field), giving $w\,G_k(a, c, w^2)$ *exactly* for $k = 1, \dots, 4$; the Krawczyk certificate was re-done with another interval library (mpmath.iv), Theorem B with msolve, and the displacements of Theorem C with a separate validated integrator (polar coordinates, angle as time), which reproduces the enclosures of §6.4.
- **Sign lemma (§4.2) and successive perturbations (§6)**: classical; the proof given is self-contained. Check that every perturbed parameter stays on $\mathcal{T}_0$ when Lemma 4.2 is used (it does: $\Phi$ maps into $\mathcal{T}_0$), and that the last step only changes $K$.
- **Interval arithmetic**: Arb (python-flint 0.9.0); comparisons used are the certain ones (< true only if true for all points of the balls). Polynomials are evaluated term by term (no Horner), which only enlarges enclosures.
- **Mass-action interpretation**: exponents are real (generalized mass-action, real complexes), exactly as in the paper's setting (§5: $a > 0$, $b > -1$, $d > 0$, $1 + bd > 0$).
- **Not proved**: exact number (=4) of small cycles and their hyperbolicity (classical, cited in the Remark of §2; for Theorem C, more cycles could in principle hide in the intervals — at least four are proved); the more precise cycle locations of §8.2 (numerical only); anything about large-amplitude cycles.

## 10 Reproduction

From the project root (Git Bash): `outils/wsl "cd chemistry-open-problems/problems/boros-hofbauer-four-limit-cycles && bash scripts/run_all.sh"`. Individual scripts: `focal_exact.py` (exact $\tilde{V}_k$), `crosscheck.py`, `certify.py`, `uniqueness.py`, `validated_cycles.py` (each prints PASS/FAIL and exits non-zero on failure; seconds each), `polar_check.py` (~5 s), `illustrate_cycles.py` (~3 min, illustration). Full log: `certificates/run_all.log` (35 PASS, 0 FAIL).

## 11 $g_1, g_2, g_3$ ($G_k = \kappa_k a^{2k} c^{2k} g_k$, $\kappa = (-\frac{1}{8}, \frac{1}{1152}, -\frac{1}{442368}, \frac{1}{530841600})$)

```
g_1 = 2*a^2*c*s^2 + 2*a^2*c*s + a*c^2*s - 2*a^2*s^2 - 4*a^2*s - a*c*s - 2*a^2 - a*c
+ c^2
```

```
g_2 = 318*a^4*c^3*s^5 + 396*a^4*c^3*s^4 + 159*a^3*c^4*s^4 - 954*a^4*c^2*s^5 + 78*a^4*c^3*s^3
+ 35*a^3*c^4*s^3 - 1998*a^4*c^2*s^4 - 765*a^3*c^3*s^4 + 954*a^4*c*s^5 - 1374*a^4*c^2*s^3
- 1086*a^3*c^3*s^3 - 3*a^2*c^4*s^3 + 3048*a^4*c*s^4 + 1053*a^3*c^2*s^4 - 318*a^4*s^5
- 330*a^4*c^2*s^2 - 409*a^3*c^3*s^2 - 79*a^2*c^4*s^2 + 3804*a^4*c*s^3 + 3391*a^3*c^2*s^3
+ 72*a^2*c^3*s^3 - 1446*a^4*s^4 - 447*a^3*c*s^4 + 2280*a^4*c*s^2 + 3511*a^3*c^2*s^2
+ 900*a^2*c^3*s^2 - 175*a*c^4*s^2 - 2748*a^4*s^3 - 2340*a^3*c*s^3 - 69*a^2*c^2*s^3
+ 570*a^4*c*s + 1173*a^3*c^2*s + 636*a^2*c^3*s + 33*a*c^4*s - 2748*a^4*s^2 - 4134*a^3*c*s^2
- 743*a^2*c^2*s^2 + 175*a*c^3*s^2 - 1446*a^4*s - 3036*a^3*c*s - 1151*a^2*c^2*s +
438*a*c^3*s - c^4*s - 318*a^4 - 795*a^3*c - 477*a^2*c^2 + 159*a*c^3 + 159*c^4
```

`g_3`: 136 terms, see `certificates/G_3.json` (and `g_4`: `certificates/G_4.json`). The JSON format is `{"vars": ["a","c","s"], "terms": [[e_a, e_c, e_s, "p/q"], ...]}` and stores $G_k$ itself (with the constant and the monomial factor).